

⋮ Hide menu

Course Introduction

What is a Linear Program?

Tutorial: Solving Linear Programs Using PuLP Library

Network Flow Problems and Linear Programming

Geometry of Linear Programs

Algorithms for Linear Programs: A brief overview

✓

Video: Algorithms for Solving Linear Programs

18 min

📖

Lab: Interactive Notes on Geometry of Linear Programs and Simplex Algorithm

1h

📝

Quiz: LP Algorithms

30 min

Problem Set # 1

LP Algorithms

Review Learning Objectives

✔ Submit your assignment

Due

Feb 11, 11:59 PM IST

✔ Receive grade

To Pass

80% or higher

👍 Like

👎 Dislike

📄 Report an issue

✔ Congratulations! You passed!

Grade received 100%

Latest Submission Grade 100%

To pass 80% or higher

Go to next item

1. Select all options that are true about vertices of a polyhedron.

1 / 1 point

✔ A vertex of a polyhedron satisfies all the constraints.

✔ Correct

Correct:

Try again

✔ Vertex lies on at least n faces of the polyhedron , where n is the number of decision variables.

✔ Correct

Correct:

Your grade

100%

View Feedback

We keep your highest score

❑ Total number of vertices in a polyhedron is always linear in the number of constraints.

❑ If any set of d faces intersect at a point for a d -dimensional polyhedron, it defines a vertex of the polyhedron.

2. Which of the following statements are true about the simplex algorithm in LP ?

1 / 1 point

✔ At every iteration , the algorithm moves to an adjacent vertex to improve the value of the objective function.

✔ Correct

Correct: It moves to an adjacent vertex only if it optimizes the objective function in comparison to the current value.

❑ The algorithm starts at the center of the feasible region and then expands towards the vertices.

✔ The average case time complexity for simplex algorithm is polynomial time.

✔ Correct

Correct:

❑ The algorithm works only for LP problems with 2 decision variables.

